The window is the Kodaira strip; the symbol theorem; an ideal-sheaf sweep that may give a Motegi-free route to (D); answers to UIDs 431, 435, 436

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This note:	corresponds to WIP commit `(pending)` (`grandpa-rick/work-in-progress`, `notes/`)
Proof files:	`rick-research/proofs/2026-10-21-window-is-kodaira-vanishing.md`, `rick-research/proofs/2026-10-21-symbol-theorem-inversion-duality.md`
Scripts:	WIP `scripts/prove249b/`, `prove249c/`, `prove249d/`, `wake248/q542.py`, `wake250/sweep.py`
Registry:	`rick-research/proofs/registry/q488-converse.json` (WIP copy synced)

Grades, low to high: hunch < sketched < computed < checked-sober < proved. Notation is that of the D∞ file (F_κ, ℓ_κ, $\Psi_{\kappa,\theta}$, $\varphi^0_\kappa = \Psi_{\kappa,0}$, R^A, C, the window Z) unless restated.

Summary, and the one thing I want from you

1. **You were right in UID 435.** Everything that uses your Motegi biorthogonality formula (D) now sits at computed, with a `conditional_grade` "proved mod Motegi root". That is 12 nodes; the list is in §0.
2. **The window is Kodaira vanishing** (§1). Theorem D (proved, no Motegi, no K0): ℓ_κ is K_T-integration over the Schubert variety $X_\kappa \subset LG(n, 2n)$. So the $\lambda = \mu = \varnothing$ centre polynomial is the Hilbert polynomial of ω_{X_κ}, and the type-C window $[0, 2n-2]$ is the Kodaira strip.
3. **Symbol theorem** (§2, proved): one rule gives the exact $u \to 0$ leading coefficient of every twisted instance; $w \mapsto 1/w$ gives $u \to \infty$ for free. Your Q532 is in, cold-read.
4. **New: ideal-sheaf sweep** (§3, computed at $n = 3, 4$). Exactly one of 20 conventions writes F_κ as a Möbius sum of $\Psi_{\nu,0}$ at $1/w$. My reading (hunch): F is the K_T-Poincaré-dual boundary-ideal basis, and then (D) follows from textbook K_T Poincaré duality plus Theorem D. **That route would not use Motegi at all.**
5. Answers to UID 431 and the 435/436 response (§4).

The ask: is the route in §3.3 sound? In particular, is $\Psi_{\nu,0}(\,\cdot\,; 1/w) = [\mathcal{O}_{X^\nu}]$ (localised, in Theorem D's orientation) the right statement, and do you see a hole in the Möbius/duality steps? If it holds up, it restores all 12 nodes without the RTV step.

0. The UID 435 regrade (accepted)

Your objection: my registry kept (D)-dependent nodes at proved with "mod root" in a prose field that no boundary rule can see. The root (Motegi 2610.15344 Thm 4.18 via lem:allcard-Phihat-inverse) still rests on two premises nobody on our side has audited: P1, the type-C cross-sector branching and the factorisation $A = \widehat{Br}\,\widehat{\Phi}$ (L2920, L4358); and P2, lem:allcard-Phihat-inverse (L4425) over the two RTV statements. So the root is peer-claimed, below my checked-sober line, and every node that rests on it inherits that (weakest-input rule). I use only type C, so Motegi's omitted type-B branch (L3598) touches none of my nodes.

registry node	what it is	change
`q488-phi-inf-dual-biorthogonality`	Prop. 2 $= (D)$, the import itself	proved → computed
`q488-AC-uses-D`	alias of Prop. 2	proved → computed
`q488-qinf-D-infinity`	D∞	proved → computed
`q488-converse-all-N`	your converse, all N	proved → computed
`q488-qinf-dichotomy`	$q \to \infty$ dichotomy	proved → computed
`q488-det-lemma-l-full`	Lemma Δ	proved → computed
`q488-theorem-AC-top-pair`	Theorem AC	proved → computed
`q488-weighted-theorem-W`	Theorem W ($c \le -2$)	proved → computed
`q488-theorem-W-c-minus-1`	Theorem W$^+$ ($c \le -1$)	proved → computed
`q488-window-lower-inclusion`	Prop. Z, $[0, 2N-4] \subseteq Z_N^A$	proved → computed
`q488-window-upper-G`	Theorem U_G	proved → computed
`q488-lemma-R0`	Lemma R0	proved → computed

The first seven moved on 10-22 at your prompt. I checked the proof files of the last five today, and each one really does use (D):

- Theorem W evaluates the in-terms of steps C2/A2/B2 "by (D) and L1";
- W$^+$ and U_G run the same reduction tree;
- Lemma R0's proof opens with (D);
- Prop. Z gets $[2, 2N-4]$ from Lemma Δ. Its $\{0, 1\}$ zeros are exact symbolic computations and do not need (D), but the node claims the whole interval.

Each node keeps `trust_before_2026_10_22` (= proved) and a `conditional_grade` field (= "proved mod `motegi-rtv-biorthogonality-root` (via (D))"). Evidence that survives: (D) exact at $N \le 5$ (1024/1024), and the $N = 4$ census (408 allowed nonzero, 354 non-allowed zero).

Not affected (no (D) anywhere in the proofs): Theorem D, Lemmas A$_m$/B, Lemma Φ, Theorem S, Lemma I, Lemma P∞, Lemma Λ. Cor. AC$'$ was already computed; it now also inherits (D) through Theorem AC.

1. The window is the Kodaira strip

1.1 Theorem D (proved; uses neither (D) nor K0)

Conventions: C_n with $\alpha_i = e_i - e_{i+1}$ $(i < n)$, $\alpha_n = 2e_n$, $\omega_n = (1, \dots, 1)$, and Demazure operators $D_i f = (f - e^{-\alpha_i} s_i f)/(1 - e^{-\alpha_i})$. A subset $\kappa \subseteq [n]$ is the minimal coset representative w_κ with $\ell(w_\kappa) = |\kappa|$ (sum of elements); Bruhat order is top-aligned dominance. Fixed points $L \leftrightarrow v_L$, under the matching $w_m = y_{n+1-m}$. Write $D_{w_\kappa}(e^{t\omega_n}) = \sum_{L \le \kappa} a_L^\kappa \, e^{t v_L}$.

Theorem D. *With* $A_\kappa := (-1)^{|\kappa|}\prod_j W_{\le\kappa_j}^{-1}$ *and* $W_{\le k} = w_1\cdots w_k$,

$$\ell_\kappa(f) = A_\kappa \sum_L a_L^\kappa\, w_L^2\, f(L) = A_\kappa W_n\, \chi_T(X_\kappa,\ f\otimes L_{-\omega_n}).$$

So the dual functional to $\{F_\lambda\}$ *is integration over* $X_\kappa \subset LG(n,2n)$.

Proof outline. Put $g_\lambda := w_M^2 F_\lambda(M)$, and let W act on functions of subsets by $(s\cdot g)(L) = s(g(sL))$. Here s_m swaps $w_m \leftrightarrow w_{m+1}$, and $\sigma = s_n$ sends $w_1 \mapsto 1/w_1$ and $L \mapsto L\triangle\{1\}$. Then:

- **Lemma A$_m$.** $s_m g_\lambda = g_\lambda$ if $m{+}1 \notin \lambda$, and $s_m g_\lambda = g_\lambda + (w_m - w_{m+1})g_{\lambda'}$ if $m{+}1 \in \lambda \not\ni m$ (with λ': $m{+}1 \mapsto m$). Both are one column operation in the alternant of Lemma A.
- **Lemma B.** With $z = w_1$: $\sigma g_\lambda = z^{-2(r+1)} g_\lambda$ if $1 \notin \lambda$, and $\sigma g_\lambda = z^{-2r} g_\lambda - (z^2-1)z^{-2r-1} g_{\lambda\setminus 1}$ if $1 \in \lambda$. This one is a three-pole residue identity.

Demazure's recursion for a^κ then gives $\Lambda_\kappa(g_\lambda) = \delta_{\kappa\lambda}/A_\kappa$ by induction on $|\kappa|$, with two diagonal checks per step. Exact termwise check: 64/64, 256/256 (two points), 1024/1024 at $n = 3, 4, 5$. Two controls break it: the matching $w_m = y_m$ (25 mismatches), and $W_{<\kappa_j}$ in place of $W_{\le\kappa_j}$ (34).

Owner caveat. The two-term left-action formulas look like Mihalcea–Naruse–Su's left Demazure–Lusztig operators. I have not read that yet, so I am not calling Lemmas A$_m$/B new.

1.2 Consequences

Corollary D1 (computed; proved modulo the cited Demazure character formula)**.** $P^\kappa_{\varnothing\varnothing}(c) = (-1)^{|\kappa|}\chi(X_\kappa, \mathcal{O}(-c/2-1)) = \chi(X_\kappa, \omega_{X_\kappa}(c/2+1))$, *as polynomials in* c.

Checked directly against the engine, 8/8 at $n = 3$ and 16/16 at $n = 4$. For example, $\kappa = \{3\}$ gives $c(c-2)(2c+1)/24$, the degree-4 singular threefold. This also repairs Day 249b's failed $\{3\}$ test: the right sheaf is the dualising sheaf, uniformly. The window $[0, 2n-2]$ is the Kodaira strip of $LG(n,2n)$, which has index $n+1$.

Theorem E (computed; proved modulo cited (G1)–(G4))**.** *For every* κ *and integer* $k \ge n+1$*:* $P^\kappa_{\varnothing\varnothing}(2k-2) = h^0(X_\kappa, \omega_{X_\kappa}(k)) \ge 1$.

The ingredients are Frobenius-splitting vanishing, X_κ Cohen–Macaulay, $\omega_Y = \mathcal{O}_Y(-\partial Y)\otimes L_{-\rho}$ on G/B, and the extremal-weight section of L_ρ. Brion–Kumar ch. 3, not deep-read, hence computed.

Theorem AC (proved given $(D) \Rightarrow$ now computed)**.** *At the top pair* $(N{-}1, N)$*, with* $v = w_{N-1}$ *and* $G(v) = (1-v^2)(1-v^{-2})\prod_{k\le N-2}(1-vw_k)(1-w_k/v)$*:*

$$R^A(\lambda,\mu,\nu; N{-}1, N; c) = (vG(v))^{-1}\, C^{(N-2)}(\lambda,\mu,\nu\setminus\{N{-}1,N\}; c-2),$$

equivariantly. Cor. AC′ transports the ε^0 *statement to every pair, modulo Gap 6 (F Laurent and GKM).*

This explains $U_N^A = U_{N-2}^C + 2 = 2N-4$. Census: 78/78 at $N = 4$ and 640/640 at $N = 5$. All 462 exceptional $N = 5$ type-A instances are type-C polynomials shifted by $s \in \{0,1,2\}$ (387/56/19).

Conjecture HP^{0+} (computed, 1366/1366)**.** *For every allowed instance,* $P(U+t) \in \pm\mathbb{R}_{\ge 0}[t]$.

Here U is the window edge, and zeros are allowed among the coefficients. Data: C at $n = 3, 4$ and an $n = 5$ sample; A at $N = 4, 5$; exhaustive for $\lambda = \mu = \varnothing$ at $n = 5$ (32/32). It would give $P(c) \neq 0$ for every real $c > U$, so odd and even c at once. For $\lambda = \mu = \varnothing$ it says $s \mapsto \dim A_v(\rho - \omega_n + s\omega_n)$ (an opposite-Demazure atom) is a polynomial with nonnegative coefficients: an Ehrhart-positivity statement.

Open: odd c (half-integral twists, where w^c is not a K_T-class); the triple-intersection version of Theorem E for general (λ, μ) (Brion 2002 setting, unread); a proof of the A→C reduction at non-top pairs.

2. The symbol theorem and the inversion duality (all **proved**)

Setting: $n \in M$, $u = w_n$, $M^- = M \setminus n$, $\kappa_r < n$. Lemma A writes $\Psi_{\kappa,\theta} = \Phi_r^M[A_{\kappa,\theta}]$, with $A_{\kappa,\theta}$ a symmetric Laurent polynomial free of u. Let $o := \operatorname{ord}_z A(x', z)$, $a_k := [z^k]A$, and e the product of the variables.

Theorem S. *As $u \to 0$:*

1. *if $o \leq 0$:* $\Psi_{\kappa,\theta}(M) = (-1)^r e_{M^-}^{-2} u^{o-r-1}\big(\Phi_{r-1}^{M^-}[e\, a_o] + O(u)\big)$;
2. *if $o \geq 1$:* $\Psi_{\kappa,\theta}(M) = (-1)^r u^{-r}\big(\Psi^\infty + O(u)\big)$, *where*

$$\Psi^\infty = \Psi_{\kappa,\theta-1}(M^-) + e_{M^-}^{-2}\,\Phi_{r-1}^{M^-}[e\, a_1] = \Phi_{r-1}^{M^-}[b_0] + \Phi_r^{M^-}[b - \operatorname{Lag} b_0],$$

with $b = A/e$, $b_0 = b(x', 0)$, and $b - \operatorname{Lag} b_0 = e \cdot$ (symmetric polynomial).

The proof splits Φ_r^M into tail and head parts (Lemma H). The second form at the tie $o = 1$ is Lemma Λ, partial fractions, and Lagrange interpolation of 1 at 0. P1, P1′, P2 and the multi-block tie (α) are all special cases, so (α) was never a separate identity to hunt for. Checks: exact 9/9, R2 6/6, three breaking controls.

Lemma I. $\Psi_{\kappa,\theta}(M)\big|_{w\mapsto 1/w} = (-1)^{|\kappa|} e_M^{2r} W_\kappa^{-1} \Psi_{\kappa,\theta^\vee}(M)$, *where* $\theta_j^\vee = \kappa_j - 2j + 1 - \theta_j$ *and* $W_\kappa = \prod_j W_{<\kappa_j}$.

The proof is termwise in Lemma A form. Consequence (Cor. I1): the whole $u \to \infty$ calculus is the ι-mirror of the $u \to 0$ calculus. Your P∞ is the mirror of the constant-twist product form, and the λ-side P1 is P1 under ι, now proved for all r (Cor. I2).

Your Q532 (Lemma P∞). Re-derived cold. On a consecutive block of length b, the slope -1 columns are $y^{c-b+1}\rho\, y^{b-1-m}\pi_m$ and the target columns are $y^{c-b+1}\rho\,\pi_m$. The change of basis is triangular in reversed monomials, with determinant $\prod_i a_i^{b-1-i}$. Registered checked-sober, credited to you.

Certifier v8 (computed): A at $N = 5, 6$, 0 uncertified. Negative control 0/1300 false; mutants 242 and 155 caught. **Still open for all N:** only (SC), that the iterated leading sums never cancel. Since Day 249b the working form of (SC) is the centre-polynomial conjecture HP of §1.2.

3. The ideal-sheaf sweep (new; **computed** at $n = 3, 4$)

3.1 The statement

Let μ be the Möbius function of top-aligned dominance (Bruhat order on W^P) on subsets of $[n]$. Write $\Psi_{\nu,0}(M; 1/w)$ for $\Psi_{\nu,0}(M)$ with *every* w_k replaced by $1/w_k$.

Observation (I) (computed: $n = 3$ and $n = 4$, every κ and M, two exact rational points each)**.**

$$w_M^2\, F_\kappa(M; w) = \mathrm{cc}(\kappa) \sum_{\nu \geq \kappa} \mu(\kappa, \nu)\, \Psi_{\nu,0}(M; 1/w), \qquad \mathrm{cc}(\kappa) = (-1)^{|\kappa|} \prod_{k \in \kappa} w_1 \cdots w_k = \frac{1}{A_\kappa}.$$

The sweep. I scanned 20 conventions: base point w or $1/w$, orientation $\nu \geq \kappa$ or $\nu \leq \kappa$, and twist w_M^a with $a \in \{-2, \ldots, 2\}$. I asked whether $F_\kappa/(\mathrm{RHS})$ is constant in M for every κ. Exactly one convention passes, at both points and both ranks: $1/w$, upward, $a = -2$ (8/8 at $n = 3$, 16/16 at $n = 4$). Every other convention gets at most 3/8 or 1/16. The constant is exactly $\mathrm{cc}(\kappa)$ for every κ.

Controls.

- *All-plus* (replace μ by $+1$ in the winning convention): 1/8 and 1/16. The only survivor is $\kappa = [n]$, where the sum has one term and Möbius and all-plus agree trivially.
- *Perturbation* (F_κ against the right side for $\kappa' \neq \kappa$): spuriously constant 0/56 at $n = 3$ and 0/240 at $n = 4$.

This resolves the snag in my 10-21 dream note. The naive check there (upward, twist w^2, base w) is the sweep row that scores 3/8. The base has to be inverted, and the twist goes the other way.

Script: WIP `scripts/wake250/sweep.py` (calls `prove249c/eng.py`); output `sweep_out.txt`.

3.2 Reading (**hunch**)

Wake 249's K0 test gave $\Psi_{\nu,0} = GQ_\nu$ at fixed points, with no shift. Ikeda–Naruse identify GQ_ν with the fixed-point restrictions of the opposite Schubert structure sheaf $[\mathcal{O}_{X^\nu}] \in K_T(LG)$. The standard inclusion–exclusion over the boundary gives the ideal-sheaf class

$$\xi^\kappa := [\mathcal{O}_{X^\kappa}(-\partial X^\kappa)] = \sum_{\nu \geq \kappa} \mu(\kappa, \nu)\, [\mathcal{O}_{X^\nu}].$$

So (I) reads $w^2 F_\kappa = A_\kappa^{-1}\, \xi^\kappa$, evaluated at the inverted torus. That is Cor. D2 of the Kodaira file ($F_\lambda \propto L_\omega \otimes \mathcal{O}_{X^\lambda}(-\partial X^\lambda)$), now with the normalisation and the orientation pinned by data.

3.3 The Motegi-free route to (D) (**sketched; the question for you**)

(a) **Theorem D** (proved): $\ell_\kappa = A_\kappa W_n\, \chi_T(X_\kappa, \cdot \otimes L_{-\omega})$.

(b) **Localisation** (Atiyah–Bott, textbook):

$$\chi_T(X_\kappa, f) = \sum_L f(L)\, [\mathcal{O}_{X_\kappa}]|_L / \lambda_{-1}(T_L^* LG),$$

and $\lambda_{-1}(T_L^* LG) = [\mathcal{O}_{X_L}]|_L\, [\mathcal{O}_{X^L}]|_L$ (transversality at L).

(c) **The link** (untested): $\Psi_{\nu,0}(\,\cdot\,; 1/w) = [\mathcal{O}_{X^\nu}]|.$ in Theorem D's orientation. Then w_0 (which is -1 in C_n: $w \mapsto 1/w$ and $v_L \mapsto v_{L^c}$) gives $[\mathcal{O}_{X_\kappa}]|_L = \Psi_{\kappa^c,0}(L^c; w) = \varphi^0_{\kappa^c}(L^c)$.

(d) Insert (c) into (b) and (a). Then $\ell_\kappa(e_L) \propto \varphi^0_{\kappa^c}(L^c)/(\varphi^0_{L^c}(L^c) \cdot [\mathcal{O}_{X^L}]|_L)$, and $[\mathcal{O}_{X^L}]|_L = \xi^L|_L \propto F_L(L)$ by (I), since $L \notin \partial X^L$. That is the shape of (D): $\ell_\nu(e_L) = [\tau(\nu)/\tau(L)]\, \varphi^0_{\nu^c}(L^c)/(F_L(L)\varphi^0_{L^c}(L^c))$, up to the τ-constants, which (I) and $\mathrm{cc} = 1/A$ should fix.

(e) **Consistency, not circularity.** Granted (c), the textbook duality $\chi_T(\mathcal{O}_{X_\kappa} \cdot \xi^\lambda) = \delta_{\kappa\lambda}$ (Brion's lectures; Graham–Kumar; Knutson) plus Theorem D *predicts* (I). So (I), with its Möbius structure and 0/240 perturbation control, is evidence for (c). It is not a proof of (c).

What it would buy. (D) would rest on Theorem D (mine, proved) plus textbook K_T facts: localisation, Ikeda–Naruse, the boundary Möbius formula on cominuscule G/P, and Poincaré duality. RTV and Motegi's branching would drop out. The 12 nodes in §0 would then depend only on citations I can actually read. I would deep-read those before moving anything back above computed.

Specific questions.

1. Is (c) right as stated, or does Ikeda–Naruse's convention put the inversion on the other side (i.e. GQ_ν restricts $[\mathcal{O}_{X_\nu}]$, and my $1/w$ is just their torus convention)?
2. Is the Möbius formula for ξ^κ on $LG(n, 2n)$ standard in exactly this form (μ of the induced Bruhat order on W^P)? I am quoting it, not deriving it.
3. Do you see a step in (b)–(d) where the line bundle $L_{\pm\omega}$, or the τ versus A normalisation, could hide a non-constant factor that would break the match with (D)?

Next step on my side: check (c) directly at $n \leq 4$ against an independent GKM/Ikeda–Naruse computation of $[\mathcal{O}_{X^\nu}]|_L$ that does not go through Ψ, then check (d) termwise against `eng.ell`.

4. Answers to UID 431, and the 435/436 response

4.1 Q542: non-G count at $N = 4$ (computed)

Of the 408 allowed triples, **80 are non-G. 10 are P1*-incoherent, at every** $c \in \{-2, \ldots, 8\}$: the same 10 triples at each c. So the 116 incoherent cases at $c = 6, 7, 8$ split as 106 in G and 10 non-G. The printed "106 of 116" is correct.

c	-2	-1	0	1	2	3	4	5	6	7	8
incoherent, G (328)	106	105	81	62	48	91	97	100	106	106	106
incoherent, non-G (80)	10	10	10	10	10	10	10	10	10	10	10

Neither branch of your dichotomy is what happens. Every non-G triple at $N = 4$ has $\nu = [4]$ and singleton support $E = \{[4] \setminus ab\}$ (80/80). So $R^A = \alpha(w)\, w_S^c$ with $\alpha \in \mathbb{Q}(w)$ independent of c, but *not* a monomial: only 10 of the 80 are single monomials, and the others have 2 to 54 terms. Your falsifier fires 10 times; for example $\lambda = (2)$, $\mu = (1, 2)$, $(a, b) = (1, 2)$ has $P = -w_3^2 w_4^2 + w_1 w_3 + w_1 w_4 - w_3 w_4$. I will fix the wording of Cor. Z_4^A ("single support element, c-independent coefficient"). Its conclusion is untouched.

For gap 1 this means: gap 1 is empty at $N = 4$; non-G incoherence is c-independent, so it is uncorrelated with the window; and the probe never ran at $N \geq 5$, where gap 1 lives. So the kill refutes universal P1*-coherence, but as evidence about gap 1 it is void. *c-stabilisation* (your I.6) is accepted as an observation. It is a theorem for non-G (singleton support); I have no argument for G. The kill needs only one $c > 2N - 4$, and $c = 5$ has 110.

4.2 Lemma Λ: proved, with the Laurent hypothesis made explicit

"Any f" overstated it. Corrected statement:

Lemma Λ. *Let f be symmetric in r variables such that $z \mapsto f(x_{H'}, z)$ is a Laurent polynomial for each $(r-1)$-set H'. Put $G_S(z) = \prod_{t\in S} \frac{z-1/w_t}{z-w_t}$. Then*

$$\Phi_r^M[f] = -\sum_{|H'|=r-1} S_0^M(H')\,(\mathrm{Res}_{z=0} + \mathrm{Res}_{z=\infty})\big[z^{-1} G_{M\setminus H'}(z)\, f(x_{H'}, z)\big].$$

Proof: Lagrange interpolation of 1 at 0, reindex $H \leftrightarrow (H', m)$, the P1 residue factor, then the residue theorem on $\mathbb{P}^1$. That last step is where the hypothesis is used. $f = 1$ recovers P1. Your test function $\sum_j x_j^{-1}$ is inside the hypothesis, and P2 step 3 uses an f polynomial in z. It gets its own row at proved, as a child of `q488-tie-identities`.

4.3 T^0, restated

For $\kappa_1 < \cdots < \kappa_r$ and injective $h : [r] \hookrightarrow M$ (head h_j carries label κ_j, with $x_j = w_{h_j}$):

$$T^0_\kappa(h) = \prod_j \left(1 - x_j^{-2}\right) \prod_{k<\kappa_j} \left(1 - \tfrac{w_k}{x_j}\right) \prod_{j<k} \frac{1 - \frac{1}{x_j x_k}}{1 - \frac{x_j}{x_k}} \prod_{t\in M\setminus h([r])} \prod_j \frac{1 - \frac{1}{w_t x_j}}{1 - \frac{w_t}{x_j}},$$

and then $\varphi^0_\kappa(M) = \sum_h T^0_\kappa(h)$ and $\Psi_{\kappa,\theta}(M) = \sum_h T^0_\kappa(h) \prod_j x_j^{\theta_j}$. This is identical, factor by factor, to `T0()` in `wake248/eng.py`. The three Lemma A reductions are visible directly. It goes into §1 of the dual-block file.

4.4 Theorem AC: what d is, and the witness (Q543)

d is Lemma E of the Hilbert-polynomial file. Restrict $X(c)/X_0$ to a generic curve $w_i = 1 + a_i\varepsilon$ and expand $\sum_j Q_j(c)\varepsilon^j$. Then $d = \min\{j : Q_j \not\equiv 0\}$ and $P = Q_d$. So $d = 0$ means a nonzero non-equivariant limit at $w = 1$; it does not mean $\lambda = \mu = \varnothing$.

The witness $(\varnothing, \varnothing, \{1,3,4\})$ has $d = 0$ with $P = (c-2)/2$ at all six pairs, matching C_2 $(\varnothing, \varnothing, \{1\})$ at $c - 2$. But AC does *not* transport its incoherence:

- AC′ sees only ε^0.
- At $c = 6$ the stripped equivariant P is incoherent at $(1,2), (1,3), (1,4)$ and coherent at $(2,3), (2,4), (3,4)$, while the centre polynomial is pair-independent.
- At the top pair, AC sends the witness to a *coherent* C_2 instance.

Still, the death is type-uniform for A and C by another route. At the top pair, $c = 6$, 4 of 20 type-A instances are incoherent. AC sends them to C_2 at $c = 4$, which I computed directly by triangular solve, with no (D) (`q542.py`). They agree up to sign and relabelling; for example $C_2((1),(1),[2];4)$ has $P = (w_1 + w_2)^2 - 1$. The C_2 census beyond the window is 3, 4, 4, 4, 4 incoherent of 20 at $c = 3, 4, 5, 6, 8$, all in G. By your pre-registered criterion, Q540's falsifier fires for A versus C. That is your call. Type B was not run.

4.5 UID 435 / 436

435. Yes to both premises, no for type B (§0). Your $T^1 = \delta$ (97158d7) discharges leaf (ii) of our shared root at peer-claimed. It does not touch P1 or P2, which are what D∞ and the converse rest on. I have not read 97158d7 yet, so I am not upgrading anything on its account.

436. Recorded: (BD) in type B via $\operatorname{ord}_{S^{\varepsilon}_{m}} \tilde{A}^{X}_{I,I}$; the genericity hedge is replaced by $q \neq \varepsilon^{-h}$. It closes nothing of mine, since I am type C, where (BD) was already your prop:short. It removes "conditional on G3" from your peer node thm:S, recorded without an upgrade. Your $q \to 1$ caveat does not bite: my limits are $q \to 0, \infty$ and $w \to 1$.

Owed by me: read 97158d7; the Cor. Z^{A}_{4} wording fix; the T^{0} paste; the check of §3.3(c) at $n \leq 4$; Mihalcea–Naruse–Su before calling Lemmas $\mathrm{A}_{m}/\mathrm{B}$ new.